

Worksheet 1A: Fractions Fundamentals

AIML Foundations Mathematics

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Part A: Numerical Fractions (12 problems)

Simplify or evaluate. Show your work.

1. $\frac{3}{4} + \frac{5}{6}$

2. $\frac{7}{8} - \frac{2}{3}$

3. $\frac{5}{9} \times \frac{3}{10}$

4. $\frac{4}{5} \div \frac{2}{15}$

5. $\frac{2}{3} + \frac{1}{4} - \frac{1}{6}$

6. $\frac{7}{12} \times \frac{8}{21} \times \frac{3}{2}$

7. $\frac{\frac{3}{4}}{\frac{9}{16}}$

8. $\frac{\frac{2}{5} + \frac{1}{3}}{\frac{1}{2}}$

9. $2\frac{1}{3} \times 1\frac{1}{2}$

10. $4\frac{2}{5} \div 1\frac{1}{10}$

11. $\frac{5!}{4!}$

12. $\frac{7!}{5! \cdot 2!}$

Part B: Pattern Sums (8 problems)

Evaluate each sum. Look for patterns.

13. $1 + \frac{1}{2} + \frac{1}{4}$

14. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

15. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

16. Based on problems 13–15, what value does this sum approach if we keep adding terms forever?
17. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4}$ (*Hint: Try writing each fraction as a difference of two simpler fractions*)
18. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5}$
19. Based on 17–18, if we kept adding terms $\frac{1}{n(n+1)}$ forever, what would the sum approach?
20. $1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!}$ *(This sum, extended infinitely, equals $e \approx 2.718\dots$)*

Part C: Algebraic Fractions — Simplification (10 problems)

Simplify completely. Assume no denominators equal zero.

21. $\frac{6p^2}{9p}$
22. $\frac{15m^3n^2}{25mn^4}$
23. $\frac{t^2 - 4}{t + 2}$
24. $\frac{r^2 - 9}{r^2 + 6r + 9}$
25. $\frac{2k^2 + 4k}{k^2 + 2k}$
26. $\frac{h^2 - 5h + 6}{h^2 - 4}$
27. $\frac{n!}{(n-1)!}$
28. $\frac{(n+1)!}{n!}$
29. $\frac{(n+2)!}{n! \cdot 2!}$
30. $\frac{6s^2 - s - 2}{2s^2 + 3s - 2}$

Part D: Algebraic Fractions — Operations (10 problems)

Perform the indicated operations and simplify.

31. $\frac{2}{p} + \frac{3}{q}$
32. $\frac{5}{t+1} - \frac{2}{t-1}$
33. $\frac{3}{m} + \frac{2}{n} - \frac{1}{mn}$
34. $\frac{r}{r+2} + \frac{2}{r-2}$
35. $\frac{1}{k-1} - \frac{1}{k+1}$
36. $\frac{2s}{s+3} \times \frac{s^2-9}{4s^2}$
37. $\frac{w^2-4}{w+5} \div \frac{w-2}{w^2+5w}$
38. $\frac{1}{u} + \frac{1}{u+1} + \frac{1}{u+2}$
39. $\frac{p^2-1}{p^2+2p} \cdot \frac{p^2+4p+4}{p^2+p-2} \div \frac{p+2}{p}$
40. If $\frac{1}{m} + \frac{1}{n} = \frac{1}{k}$, solve for k in terms of m and n .

Part E: Complex Fractions (6 problems)

Simplify each expression.

41. $\frac{\frac{1}{p} + \frac{1}{q}}{\frac{1}{p} - \frac{1}{q}}$
42. $\frac{\frac{2}{t+1} - \frac{1}{t}}{\frac{3}{t}}$
43. $\frac{1 - \frac{1}{n^2}}{1 + \frac{1}{n}}$
44. $\frac{\frac{m}{n} + \frac{n}{m}}{\frac{m^2-n^2}{mn}}$
45. $\frac{1}{1 + \frac{1}{1+\frac{1}{k}}}$
46. $\frac{\frac{1}{h} - \frac{1}{h+1}}{h}$

Part F: Repeated Multiplication — Shrinking vs Growing (6 problems)

47. Calculate each:

(a) $\frac{3}{4}$

(b) $\left(\frac{3}{4}\right)^2$

(c) $\left(\frac{3}{4}\right)^3$

(d) $\left(\frac{3}{4}\right)^4$

(e) What happens to $\left(\frac{3}{4}\right)^n$ as n gets very large?

48. Calculate each:

(a) $\frac{5}{4}$

(b) $\left(\frac{5}{4}\right)^2$

(c) $\left(\frac{5}{4}\right)^3$

(d) $\left(\frac{5}{4}\right)^4$

(e) What happens to $\left(\frac{5}{4}\right)^n$ as n gets very large?

49. Complete the rule: When you repeatedly multiply a fraction that is _____ than 1, the result approaches _____. When you repeatedly multiply a fraction that is _____ than 1, the result grows toward _____.

50. $\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4$

51. $\left(\frac{9}{10}\right)^{10}$ — estimate first, then calculate. Is it closer to 0, 0.5, or 1?

52. $\left(\frac{99}{100}\right)^{100}$ — without calculating exactly, estimate whether this is closer to 0, 0.3, 0.5, or 1. Explain your reasoning.

Part G: Basic "What Happens When..." (8 problems)

No calculation required — describe the behavior.

53. Consider $\frac{n}{100}$ where $n > 0$.

- (a) As n gets very large?
 - (b) As n approaches 0?
54. Consider $\frac{100}{n}$ where $n > 0$.
- (a) As n gets very large?
 - (b) As n approaches 0?
55. For $\frac{k+1}{k}$ where $k > 0$:
- (a) Rewrite as a sum of two fractions.
 - (b) As $k \rightarrow \infty$, what value does this approach?
56. For $\frac{t^2}{t^2+1}$ where $t > 0$:
- (a) Calculate when $t = 1$, $t = 10$, $t = 100$.
 - (b) What value does this approach as $t \rightarrow \infty$?
 - (c) Will it ever actually reach that value?
57. For $\frac{1}{1+n}$ where $n > 0$:
- (a) As $n \rightarrow 0$?
 - (b) As $n \rightarrow \infty$?
58. For $\frac{n}{n+1}$ where $n > 0$:
- (a) As $n \rightarrow \infty$?
 - (b) Is the fraction always less than 1, equal to 1, or greater than 1?
59. You have fraction $\frac{a}{b}$ where $a, b > 0$.
- (a) If you double both numerator and denominator, what happens to the value?
 - (b) If you add 1 to both numerator and denominator, does the value increase, decrease, or depend on something? *(Test with $\frac{1}{4}$ and $\frac{3}{4}$)*
60. For $\left(1 + \frac{1}{n}\right)$ where $n > 0$:
- (a) Calculate when $n = 1$, $n = 10$, $n = 100$.
 - (b) What value does this approach as $n \rightarrow \infty$?
 - (c) Bonus: $\left(1 + \frac{1}{n}\right)^n$ approaches a famous number as $n \rightarrow \infty$. Any guesses?